Bitwise Tuples

Chef has two numbers N and M. Help Chef to find number of integer N-tuples $(A_1, A_2, \ldots, A_N)$ such that $0 \le A_1, A_2, \ldots, A_N \le 2^M - 1$ and $A_1 \& A_2 \& \ldots \& A_N = 0$, where $\&$ denotes the bitwise AND operator.

Since the number of tuples can be large, output it modulo $10^9 + 7$.

Input

- The first line contains a single integer T denoting the number of test cases. The description of T test cases follows.
- The first and only line of each test case contains two integers N and M.

Output

For each test case, output in a single line the answer to the problem modulo $10^9 + 7$.

Constraints

- $1 \le T \le 10^5$
- $1 \le N, M \le 10^6$

###Subtasks **Subtask #1 (100 points):** original constraints

Sample 1:

Input	Output
4	1